TEXORA · LECTURE SHEET

CVP Analysis and Short-Run Decisions

Relevant costing in action: profit/volume charts, special orders at spare capacity, make-or-buy, optimal use of a limiting factor, and the qualitative checks that must accompany every numerical answer.

AC-4205 · COST AND MANAGEMENT ACCOUNTING · TOPIC 8 · LEVEL 4 TERM 2

1. The decision rule behind everything

A short-run decision compares **relevant costs and revenues** — future amounts that *differ between the alternatives* (Topic 2). Fixed costs that continue either way are irrelevant; sunk costs are irrelevant; opportunity costs, though invisible in the ledger, are fully relevant. Each case below is that one rule wearing a different uniform, and each is stated first as the bookkeeping trap it avoids.

2. The profit/volume (P/V) chart

The P/V chart plots profit directly against sales volume: the line starts at the fixed-cost loss (zero sales), rises at the contribution rate, and crosses zero at break-even. It is the fastest visual for "how much volume buys how much profit".

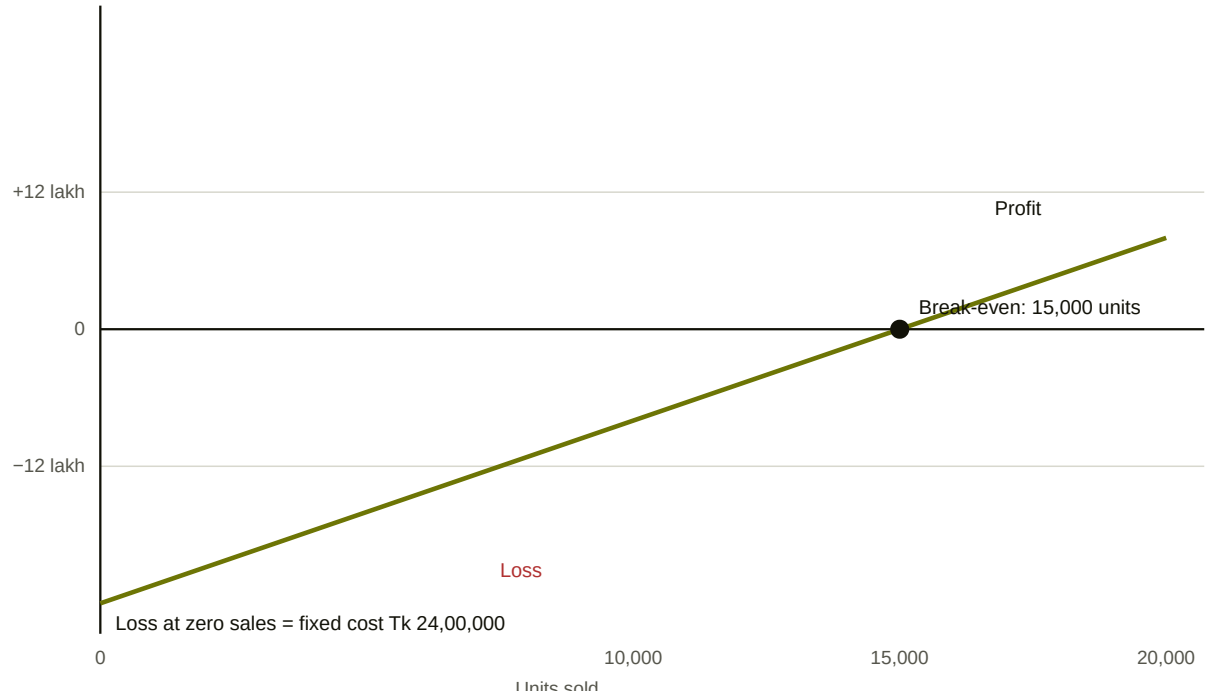


Figure 1 — Profit/volume chart for the case of Section 3 (fixed costs Tk 24,00,000, contribution Tk 160 per unit). The slope of the line is the contribution per unit; its zero crossing is the break-even point; the vertical intercept is the fixed cost.

3. Special orders — worked example

WORKED EXAMPLE — A ONE-OFF ORDER BELOW NORMAL PRICE

Capacity 20,000 units. Current sales: 15,000 units at Tk 500, variable cost Tk 340, fixed costs Tk 24,00,000.

Current contribution = (500 − 340) × 15,000 = 160 × 15,000 = Tk 24,00,000 — exactly the fixed costs: the factory sits at break-even (as the chart shows).

A buyer offers a one-off order for 3,000 units at Tk 400. Spare capacity exists (20,000 − 15,000 = 5,000), so no regular sale is lost and fixed costs do not change.

Relevant comparison per unit: price Tk 400 vs variable cost Tk 340 → extra contribution Tk 60.

Gain from accepting = 60 × 3,000 = **Tk 1,80,000 of extra profit**.

Accept — on the numbers. The full-cost view (which loads fixed cost per unit and shows a "loss" at Tk 400) is the trap: those fixed costs are paid whether the order runs or not.

Qualitative checks before signing: will the low price leak into the regular market and undermine the Tk 500 price? Is the capacity truly spare for the whole order period? Are there hidden costs (special packing, testing, air freight risk)? A special order is an exception justified by spare capacity — repeated, it becomes the price.

4. Make-or-buy — worked example

WORKED EXAMPLE — MAKING A COMPONENT VERSUS BUYING IT

A garment factory makes its own printed labels: 2,00,000 labels a year. Cost per label: direct material Tk 12, direct labour Tk 8, variable overhead Tk 5, and fixed overhead Tk 6 of which Tk 2 per unit is avoidable supervision that would disappear if production stopped (the rest is factory rent shared by all lines, continuing either way).

Relevant cost of making = 12 + 8 + 5 + 2 = **Tk 27 per label**.

A supplier offers labels at **Tk 30** per label meeting the same specification.

Decision: make. Saving = (30 − 27) × 2,00,000 = **Tk 6,00,000 per year**.

The trap avoided: comparing the supplier's Tk 30 with the full absorption cost Tk 31 (12+8+5+6) would have shouted "buy" — because Tk 4 of that fixed overhead continues regardless and must not be loaded onto the make option.

If buying frees capacity for other profitable work, that work's contribution is an opportunity benefit of buying and enters the comparison. Quality control, supply security and confidentiality of designs are the standard qualitative counterweights.

5. Limiting-factor (key-factor) decisions — worked example

When one resource is scarce — machine hours, a skilled operation, a material in short supply — profit is maximised by ranking products on **contribution per unit of the limiting factor**, not contribution per unit of product:

WORKED EXAMPLE — ALLOCATING SCARCE FABRIC

Only 6,000 kg of a fabric is available this month. Product A: contribution Tk 200 per unit, uses 2 kg → Tk 100 per kg. Product B: contribution Tk 270 per unit, uses 3 kg → Tk 90 per kg.

Ranking by contribution per unit says B (Tk 270 > Tk 200) — wrong. Ranking per kg of the scarce fabric says A first.

All fabric to A: 6,000 ÷ 2 = 3,000 units × Tk 200 = **Tk 6,00,000 contribution**.

All fabric to B: 6,000 ÷ 3 = 2,000 units × Tk 270 = Tk 5,40,000.

Choosing A earns **Tk 60,000 more** from the same fabric. (With demand limits on A, fill A's demand first, then give the residue to B.)

With several limiting factors the problem leaves arithmetic for linear programming, but the ranking principle — contribution per unit of whatever is truly scarce — remains the idea being optimised.

Glossary

Relevant cost — a future cost differing between alternatives. **P/V chart** — profit plotted against volume; slope = contribution per unit. **Spare capacity** — capacity unused by regular demand, the precondition for special-order logic. **Limiting (key) factor** — the resource whose scarcity constrains output. **Contribution per limiting factor** — the correct ranking measure under scarcity. **Avoidable cost** — cost that disappears if an activity stops; the only fixed cost relevant to make-or-buy and closure decisions.

Key points

• One rule everywhere: compare future amounts that differ; ignore sunk and unchanged fixed costs; include opportunity costs.
• Special order at spare capacity: Tk 400 against variable cost Tk 340 earns Tk 60 × 3,000 = Tk 1,80,000 — the full-cost "loss" is an illusion.
• Make-or-buy: only avoidable fixed cost joins the make side (Tk 27 vs buy Tk 30 → making saves Tk 6,00,000 a year).
• Scarcity: rank by contribution per kg/hour of the limiting factor — A (Tk 100/kg) before B (Tk 90/kg), worth Tk 60,000 on 6,000 kg.

Practice

1. A factory sells 7,000 units at Tk 800 (variable cost Tk 520, fixed costs Tk 19,60,000) with capacity 10,000. A one-off buyer offers 2,000 units at Tk 600. Should it accept, on the numbers? Quantify.
2. A component costs, per unit: material Tk 20, labour Tk 10, variable overhead Tk 8, fixed overhead Tk 9 of which Tk 4 is avoidable. A supplier quotes Tk 45. Annual volume 50,000 units. Make or buy, and by how much?
3. Products P and Q have contributions Tk 150 and Tk 240 per unit and use 3 and 6 machine hours respectively. Only 1,800 hours are available and both have unlimited demand. Which product should the hours serve, and what total contribution results?

Answer key

1. Current contribution = (800 − 520) × 7,000 = 280 × 7,000 = Tk 19,60,000 = fixed costs (break-even now). Spare capacity = 3,000 units ≥ the 2,000 offered. Extra contribution = (600 − 520) × 2,000 = **Tk 1,60,000** — accept, subject to the qualitative checks (price leakage, delivery, hidden costs).
2. Relevant make cost = 20 + 10 + 8 + 4 = **Tk 42** vs buy Tk 45 → **make**, saving (45 − 42) × 50,000 = **Tk 1,50,000 per year**. (The full absorption cost Tk 47 would wrongly suggest buying.)

3. Contribution per hour: P = 150 ÷ 3 = Tk 50/hour; Q = 240 ÷ 6 = Tk 40/hour. Serve P first: 1,800 ÷ 3 = 600 units × Tk 150 = **Tk 90,000** total contribution (Q alone would give 300 × 240 = Tk 72,000).
